



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 10

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Objective of Lab No. 10:

After performing lab 10, students will be able to:

1. Visualize the data using Matplotlib
2. Customize Visualizations

Lab Tasks:

Task 01: Histogram of Passenger Age

Create a histogram for the age column.

- Remove missing values from age
- Use `plt.hist()`
- **Set:** figure size, title, x-label and y-label, grid lines and x-ticks at suitable intervals

Task 2: Bar Chart of Passenger Count by Gender

Create a **bar chart** showing number of male and female passengers.

Add: title, axis labels, custom bar colors and values on top of each bar

Task 3: Group the data by class and calculate the average fare for each class, then plot a bar chart.

Task 4: GroupBy Embarkation Port and Plot Average Fare



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Note: Use Titanic Dataset